

Poison in the bowl: Looming health crisis from arsenic in rice

Rice, the lifeblood of Indian **cuisine**, **is** now under threat. A recent international study published in The Lancet Planetary Health warns that climate change may be **fuelling** a dangerous rise in inorganic arsenic levels in rice, potentially **triggering** a **surge** in cancer cases across Asia by 2050.

Rice is one of the earliest cereals known to humanity and especially held in great **reverence** in India, with the ancient Vedic scriptures mentioning it as annam. Today, India is a major rice producer, being the second-largest in the world after China.

In 2024, India produced nearly 138 million metric tons of rice. This significant production contributes to the food security and economic **prosperity** of the nation, besides enabling it to be a leading global rice exporter as well. Rice also happens to be a **staple food** of the nation, making it the second-largest consumer of rice globally, after China.

India **logged** an estimated consumption of 121 million metric tons of rice in the 2024–25 season, with an average **per capita** consumption of 52–55 kg per year, or 4.5 kg per person per month. Rice always has been a worry-free part of every Indian's diet due to the perception that nothing much could go wrong with its quality.

However, those carefree days seem to be over. According to a new report published in The Lancet Planetary Health on April 16, 2025, researchers from China and the US, led by Dongming Wang of the Chinese Academy of Sciences, used Free — Air CO₂ Enrichment (FACE) technology to analyse 28 rice varieties over a **decade** and discovered that **high CO₂ levels**, aided by a rise in temperatures beyond 2°C, **could trigger a spike** in the inorganic arsenic levels in rice grains. **Even brown rice**, generally considered a safer and healthier choice, **is** not safe — as in this type of rice, the outer bran layer is **intact** and **tends to** have higher arsenic levels than white rice, as the arsenic concentrates in the **bran**.

According to the World Health Organisation, arsenic **contamination** in rice must not exceed 1 mg per kg; however, in many nations, this limit has been **surpassed**. Based on the **findings**, the researchers **highlighted** that **exposure** to inorganic arsenic through rice **can** increase cancer and **associated** health risks — especially across multiple Asian nations such as Bangladesh, China, India, Indonesia, Myanmar, the Philippines, and Vietnam by 2050.

The health risk **projections**, arrived at using advanced modelling techniques, **suggested** a **surge** in cancer cases — especially lung and bladder cancers. China is expected to face the highest impact, with roughly 13.4 million additional cancer cases tied to arsenic in rice. Climate change — led **deviations** in soil composition and water are the main **drivers** of the **spike** in arsenic absorption in rice crops. **Moreover**, the **customary practice** of flood irrigation in rice cultivation also **tends** to contribute to arsenic **accumulation** in rice — especially in areas with naturally high arsenic levels in the soil or water. The study **brings to the fore** the threat **posed** to global food safety by arsenic in rice, as this staple food is the main source of nutrition for billions across the world. Left **unchecked**, arsenic levels may cause an **unprecedented** health and food security crisis. Adequate **climate adaptation** and **mitigation** strategies that are region and country-specific are urgently needed to **address** the problem. This can help design suitable interventions at appropriate levels so that the purity of rice can be maintained. India must set the **pace in this regard** so that this silent **threat** posed by arsenic **does** not become

an unmanageable health crisis by 2050. The first challenge faced by India is the lack of awareness among the general population regarding the potential presence of arsenic in rice and its **adverse** effects.

The common person purchasing loose rice from a neighbourhood grocery store or from a Government ration shop, or ordering online, may not even know what arsenic is, how it can find its way into the rice he has just bought, and what damage it can do to the human body. The general **practice** of thoroughly washing and cooking rice **can to some extent** reduce the arsenic levels but cannot **rid** the same entirely. **Given** the **pervasive** ignorance and lack of awareness regarding the presence of arsenic in rice, the Government must make it mandatory for all rice sale points in the country to follow a strict process of sharing quality — related information in clear, **legible** language. The quality information can be in the form of a Certificate of Analysis (COA) from a Government laboratory that clearly **spells out** critical quality **parameters** of the rice batch — such as level of arsenic, heavy metals and pesticide content, milling quality, grain size and shape, and nutritional content.

This information should be made available in the form of a QR code as well, so that a simple smartphone can download the COA. This **practice**, when made **widespread**, **will** begin the critical chapter of educating the consumer — who in turn can make a responsible choice. Currently, no such availability of COA is there; as a result, the consumer is assuming best quality when he makes a purchase. Similarly, the Government must put **in place** strict systems at the agricultural end, where paddy crops are being grown.

The local administration must be **equipped** with **robust** and accurate arsenic testing kits that can take spot samples of soil and water and measure the arsenic content of the same. The same data must be **relayed** back to the higher authorities so that arsenic level variation and spikes can be immediately addressed. Based on the **collation** of such data, it will also become possible to identify which parts of India have a higher incidence of arsenic in rice. This can help initiate further investigations behind this **phenomenon** and take corrective measures in the form of improving soil and water quality — the two major **determinants** for a paddy crop.

Additionally, there are many **portable** home arsenic testing kits available specifically for rice. **In order** to prevent a **calamitous** disease burden arising out of rising levels of arsenic in rice by 2050, the Government can promote the sale and usage of home arsenic testing kits for rice. These kits are generally very simple to use, and results are reasonably accurate.

This initiative can **go a long way** in preventing consumers from buying and consuming low-quality, high-arsenic contaminated rice and opt for better-quality rice. This in turn will also discourage the sellers — who, when faced with an aware consumer, will stop selling rice with high contamination levels. Rice is the **mainstay** diet for India, and arsenic contamination can have serious health and food security **ramifications** for India unless corrective measures are taken..

[Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

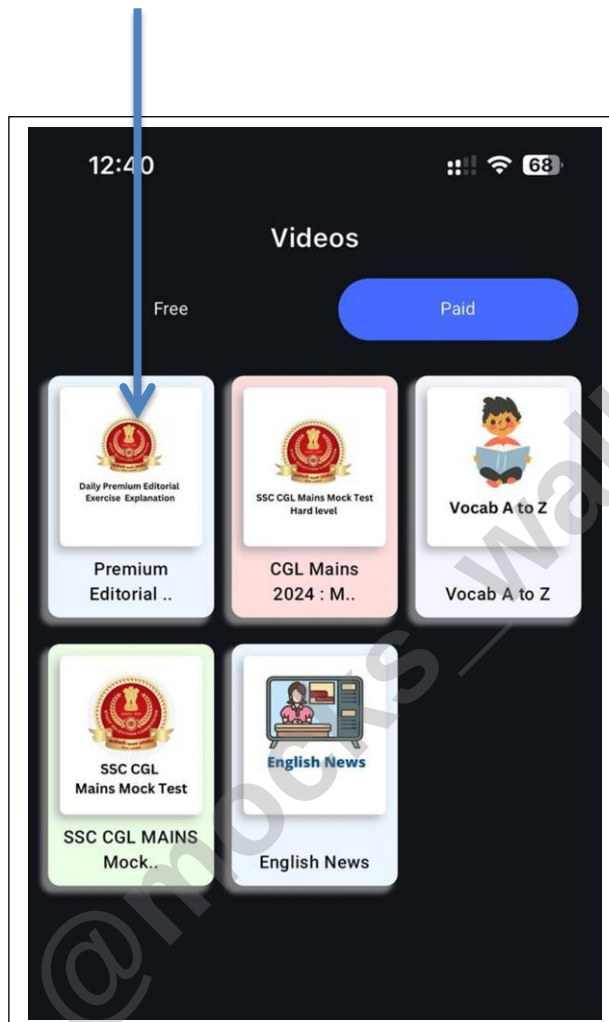
Vocabulary

1. **Looming** (adjective) – impending, approaching, threatening, imminent, forthcoming आसन्न
2. **Arsenic** (adjective) – toxic, poisonous, noxious, harmful, contaminant आर्सेनिक युक्त
3. **Cuisine** (noun) – food, cooking, gastronomy, culinary style, fare पाक शैली
4. **Fuel** (verb) – stimulate, provoke, boost, incite, intensify बढ़ावा देना
5. **Trigger** (verb) – initiate, spark, cause, prompt, generate उत्पन्न करना
6. **Surge** (noun) – increase, rise, spike, upswing, escalation उछाल
7. **Reverence** (noun) – respect, veneration, admiration, devotion, esteem श्रद्धा
8. **Prosperity** (noun) – affluence, wealth, success, growth, well-being समृद्धि
9. **Staple food** (noun) – primary food, mainstay, essential, basic diet, dietary staple मुख्य भोजन
10. **Log** (verb) – record, register, document, note, track दर्ज करना
11. **Per capita** (adjective) – per person, individual, each person, average per head, per unit प्रति व्यक्ति
12. **Decade** (noun) – Period of ten years दशक
13. **Trigger** (verb) – provoke, start, activate, instigate, initiate चालू करना
14. **Spike** (noun) – peak, surge, increase, jump, escalation तेज़ वृद्धि
15. **Intact** (adjective) – whole, unbroken, undamaged, preserved, untouched अखंड
16. **Tend** (to) (verb) – be likely, incline, be prone, have a tendency, usually झुकाव होना
17. **Bran** (noun) – the brown outer covering of grains that is left when the grain is made into flour भूसी
18. **Contamination** (noun) – pollution, tainting, impurity, adulteration, infection संदूषण
19. **Surpass** (verb) – exceed, outstrip, beat, go beyond, eclipse पार करना
20. **Finding** (noun) – result, conclusion, outcome, discovery, observation निष्कर्ष
21. **Highlight** (verb) – emphasize, underline, stress, draw attention, point out जोर देना
22. **Exposure** (noun) – contact, encounter, risk, vulnerability, subjection संपर्क
23. **Associated** (adjective) – connected, linked, related, correlated, allied संबंधित
24. **Projection** (noun) – forecast, prediction, estimate, outlook, anticipation पूर्वानुमान

25. **Surge** (noun) – increase, escalation, jump, spike, rise वृद्धि
systems to cope with the actual or expected impacts of climate change जलवायु अनुकूलन
26. **Deviation** (noun) – divergence, variation, difference, anomaly, shift विचलन
27. **Driver** (noun) – cause, force, trigger, agent, influence प्रेरक कारण
28. **Spike** (noun) – increase, surge, jump, escalation, upswing उछाल
29. **Moreover** (adverb) – furthermore, also, besides, additionally, in addition इसके अलावा
30. **Customary** (adjective) – usual, traditional, conventional, habitual, routine प्रचलित
31. **Accumulation** (noun) – build-up, collection, gathering, aggregation, amassing संचय
32. **Bring to the fore** (phrase) – highlight, emphasize, reveal, expose, showcase प्रमुखता देना
33. **Pose** (verb) – present, create, constitute, introduce, raise उत्पन्न करना
34. **Unchecked** (adjective) – uncontrolled, unrestrained, rampant, unregulated, uncurbed अनियंत्रित
35. **Unprecedented** (adjective) – unparalleled, never seen before, extraordinary, unmatched, novel अभूतपूर्व
36. **Climate adaptation** (noun) – Refers to the process of adjusting natural or human
37. **Mitigation** (noun) – alleviation, reduction, moderation, relief, control न्यूनीकरण
38. **Address** (verb) – tackle, deal with, handle, confront, attend to समाधान करना
39. **Pace** (noun) – speed, rate, tempo, stride, momentum गति
40. **In this regard** (phrase) – concerning this, in connection, on this matter, relating to this, about this इस संदर्भ में
41. **Adverse** (adjective) – harmful, unfavorable, detrimental, negative, hostile प्रतिकूल
42. **To some extent** (phrase) – partially, somewhat, in part, moderately, up to a point कुछ हद तक
43. **Rid** (verb) – eliminate, remove, clear, free, get rid of छुटकारा पाना
44. **Given** (preposition) – considering, taking into account, in view of, based on, owing to को देखते हुए
45. **Pervasive** (adjective) – widespread, omnipresent, extensive, universal, prevalent व्याप्त
46. **Legible** (adjective) – readable, clear, decipherable, intelligible, plain पठनीय

47. **Spell out** (phrasal verb) – explain, state clearly, detail, articulate, define विस्तार से बताना
48. **Parameter** (noun) – criterion, standard, limit, specification, measure मानदंड
49. **Widespread** (adjective) – extensive, prevalent, far-reaching, universal, broad व्यापक
50. **In place** (phrase) – established, implemented, in effect, operational, set up लागू
51. **Equip** (with) (verb) – provide, furnish, supply, arm, outfit सुसज्जित करना
52. **Robust** (adjective) – strong, sturdy, resilient, effective, reliable मज़बूत
53. **Relay** (verb) – transmit, pass on, communicate, forward, send भेजना
54. **Collation** (noun) – compilation, collection, comparison, assembly, gathering संग्रहण
55. **Phenomenon** (noun) – event, occurrence, incident, development, situation घटना
56. **Determinant** (noun) – factor, cause, influence, contributor, condition निर्धारक
57. **Portable** (adjective) – mobile, movable, transportable, compact, handy सुवाह्य
58. **In order to** (phrase) – so as to, with the aim to, for the purpose of, to, for ताकि
59. **Calamitous** (adjective) – disastrous, catastrophic, ruinous, tragic, devastating विनाशकारी
60. **Go a long way** (phrase) – be effective, be helpful, contribute greatly, be beneficial, make a difference बहुत सहायक होना
61. **Mainstay** (adjective) – essential, primary, key, principal, cornerstone मुख्य आधार
62. **Ramification** (noun) – consequence, result, outcome, implication, effect दुष्परिणाम

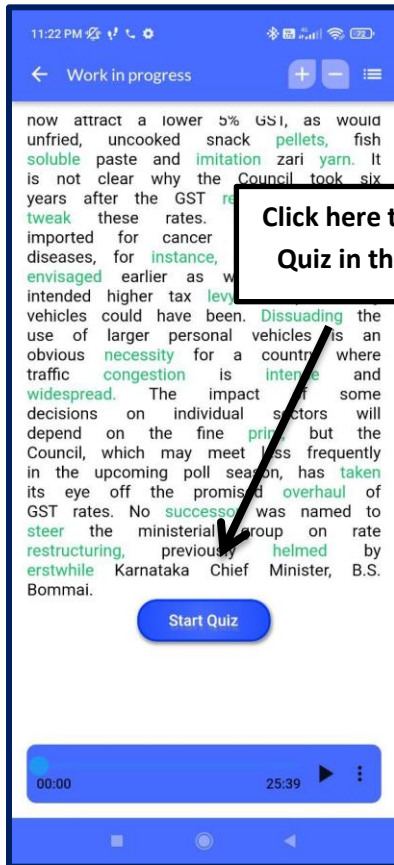
Video solution of Practice Exercise from this Editorial is exclusively available on App.



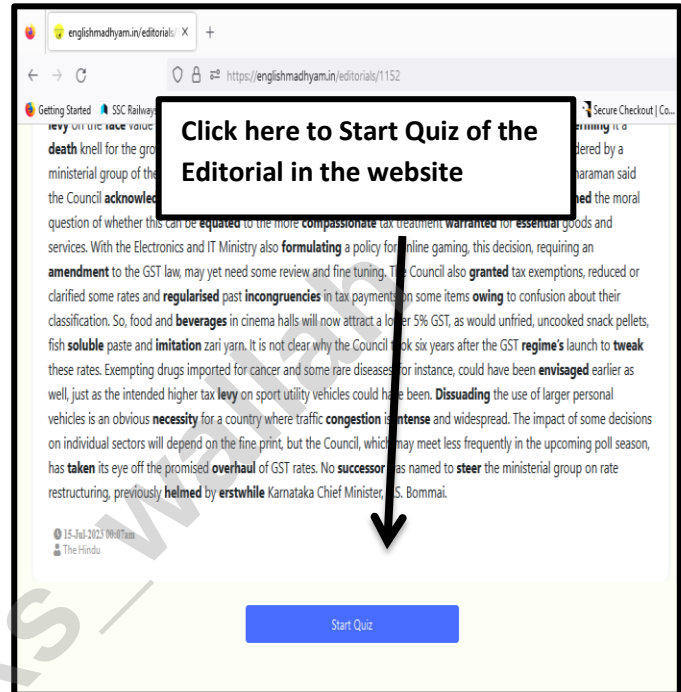
Summary of the Editorial

1. **Arsenic Threat in Rice** – A global study warns that climate change is increasing inorganic arsenic levels in rice, posing major health risks.
2. **India's Rice Dependence** – India is the second-largest producer and consumer of rice, with 138 million metric tons produced and 121 million consumed in 2024–25.
3. **Rising Cancer Risk** – Arsenic-laced rice could cause a surge in cancer cases, especially lung and bladder cancers, by 2050.
4. **High-Risk Countries** – Nations like India, China, Bangladesh, Indonesia, Myanmar, Philippines, and Vietnam are particularly vulnerable.
5. **Scientific Study** – The research, using FACE technology, found that higher CO₂ and temperatures (above 2°C) elevate arsenic in rice grains.
6. **Brown Rice Also Unsafe** – Brown rice retains more arsenic due to its bran layer, previously thought to be healthier.
7. **Irrigation Practice Worsens Issue** – Flood irrigation, common in rice farming, exacerbates arsenic uptake, especially in areas with naturally high arsenic in soil and water.
8. **WHO Safety Limit Exceeded** – Arsenic levels in rice often exceed the 1 mg/kg limit set by the World Health Organisation.
9. **Low Public Awareness** – Most Indians are unaware of arsenic contamination and its health hazards.
10. **Call for Quality Labels** – The editorial urges mandatory Certificate of Analysis (COA) at all rice sale points, highlighting arsenic and other safety parameters.
11. **Digital Access for Consumers** – Suggests integrating COA with QR codes, enabling easy access via smartphones.
12. **Need for Field Testing** – Local administrations should use arsenic testing kits for soil and water to monitor contamination in real time.
13. **Geo-Mapping Arsenic Zones** – Systematic data collection can help identify high-risk regions and inform targeted interventions.
14. **Promotion of Home Testing** – Advocates for home arsenic testing kits to empower consumers and encourage safer rice choices.
15. **Urgent Government Action** – India must lead in climate-resilient and food-safe policies to prevent a looming health and food security crisis by 2050.

Practice Exercise: SSC Pattern Based



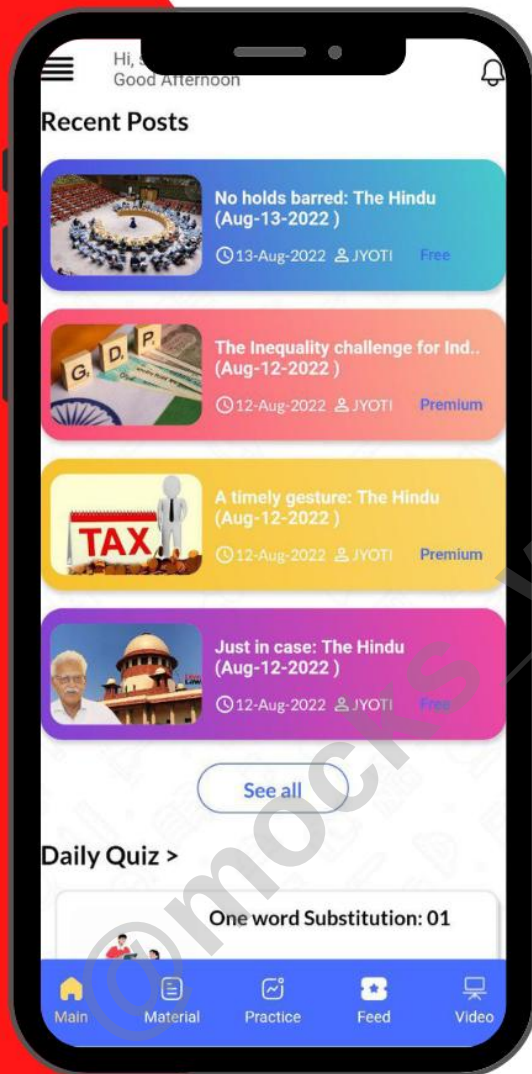
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